

How is the Weather Forecasted?

- The majority of weather forecasts are created using Numerical Weather Prediction (NWP) models
- NWP models assimilate data from multiple sources, including geo-stationary satellites, weather stations, ground-radar, radiosondes, and more, to construct a model of the current state of the atmosphere
- NWPs then compute various physics-based equations, that attempt to model the fluid dynamics of the atmosphere and oceans from the initial state to one timestep in the future. These equations are non-linear and therefore discretized across a 3D grid to derive useful results
- This model then inputs the conditions from the previously-computed timestep in order to recursively derive the desired forecast

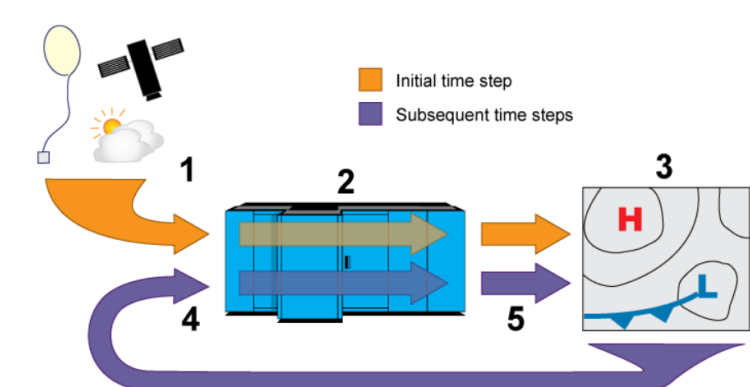


Figure 1. Simple Representation of NWP processing (NOAA 2023).

Why Weather Forecast Accuracy Decreases Over Time?

- Fluid dynamics-based equations are resolved explicitly to reduce computational cost and constraint
 - Meaning the values computed for the next timestep are approximated to a certain degree
- Additionally, more approximation occurs for atmospheric processes that are very complex and occur at scales smaller than the grid spacing. These include radiation schemes, cumulus cloud formation, cloud microphysics, and more. The estimated representation of these processes is known as parameterization.
 - Many of the processes that contribute to precipitation occur at the sub-grid scale and therefore make predicting precipitation a challenge even with all the advancements in NWPs.
- These two factors along with the computational constraint that comes with increased horizontal resolution leads to forecasts that cannot accurately model the weather with a lead time of over 7-10 days on average.
 - “A seven-day forecast can accurately predict the weather about 80 percent of the time and a five-day forecast can accurately predict the weather approximately 90 percent of the time. However, a 10-day...forecast is only right about half the time” (NOAA SciJinks, 2023).

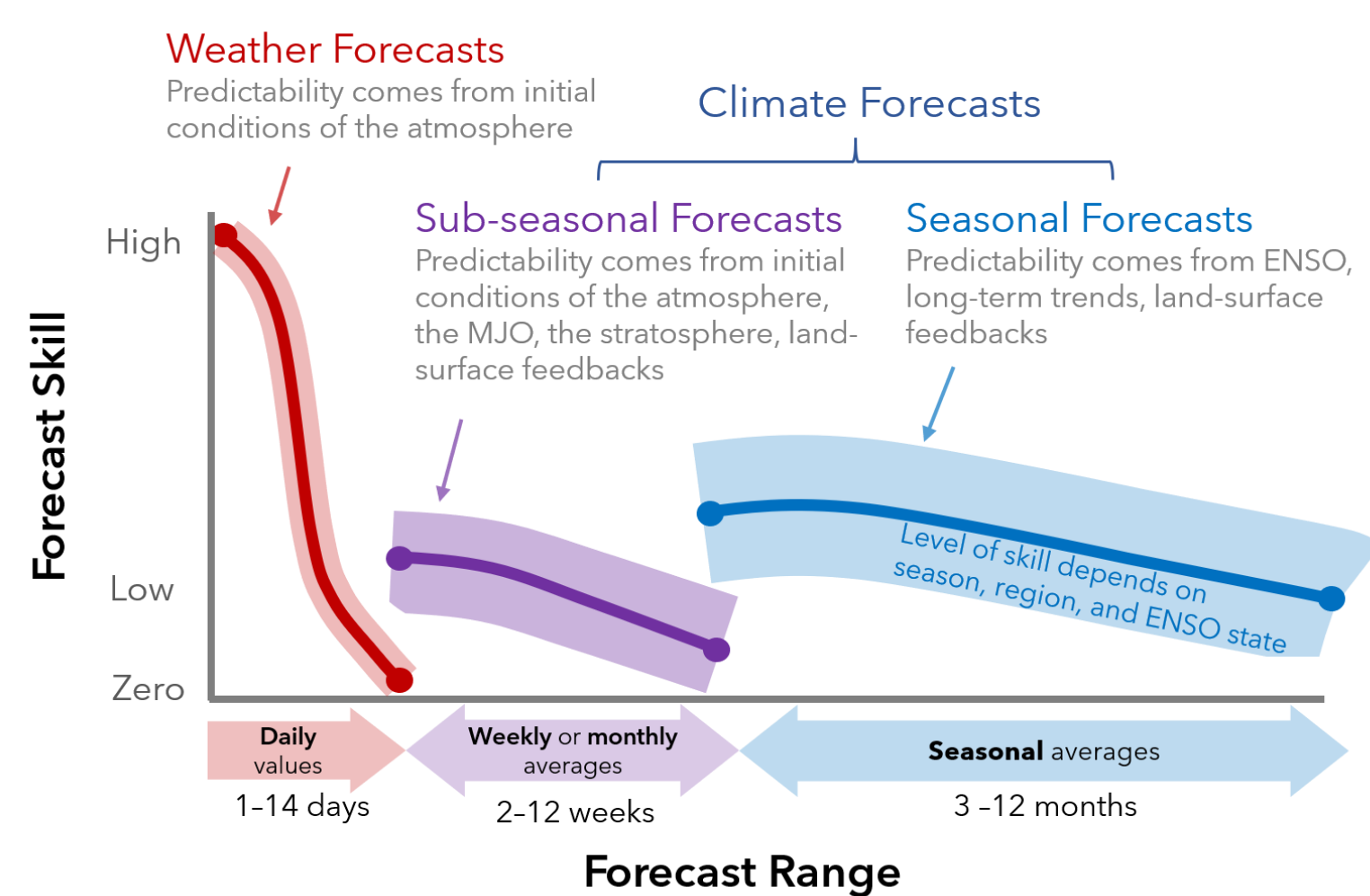


Figure 2. Forecast Skill vs. Forecast Range (lead-time) for different types of forecasts (Lukas, 2020).

Case Study: Snowstorm in the Hudson Valley on March 2, 2018.

- A major snow-storm hit the Hudson Valley on March 2, 2018.
- The National Weather Service (NWS) blended warm and cold operational forecasts to create a forecast that predicted precipitation in the region as a blend of snow and rain.
 - However, predicted snow depth in the Albany area differed from observations by approximately 8-12 inches.
- Total precipitation and snow height from the Weather Research and Forecasting Advanced Research (WRF-ARW) model output is compared to weather station observations from the NYS Mesonet weather station network. (Figure 3.)
 - 28 of the 126 NYS Mesonet station network across New York State fall within the model output domain in the Hudson Valley.
 - Total precipitation and snow height observations from 3 of the 28 stations, nearest Albany, is compared to the model output in Figure 4.
 - Selected stations (in red):
 - Voorheesville
 - Schodack
 - Schaghticoke

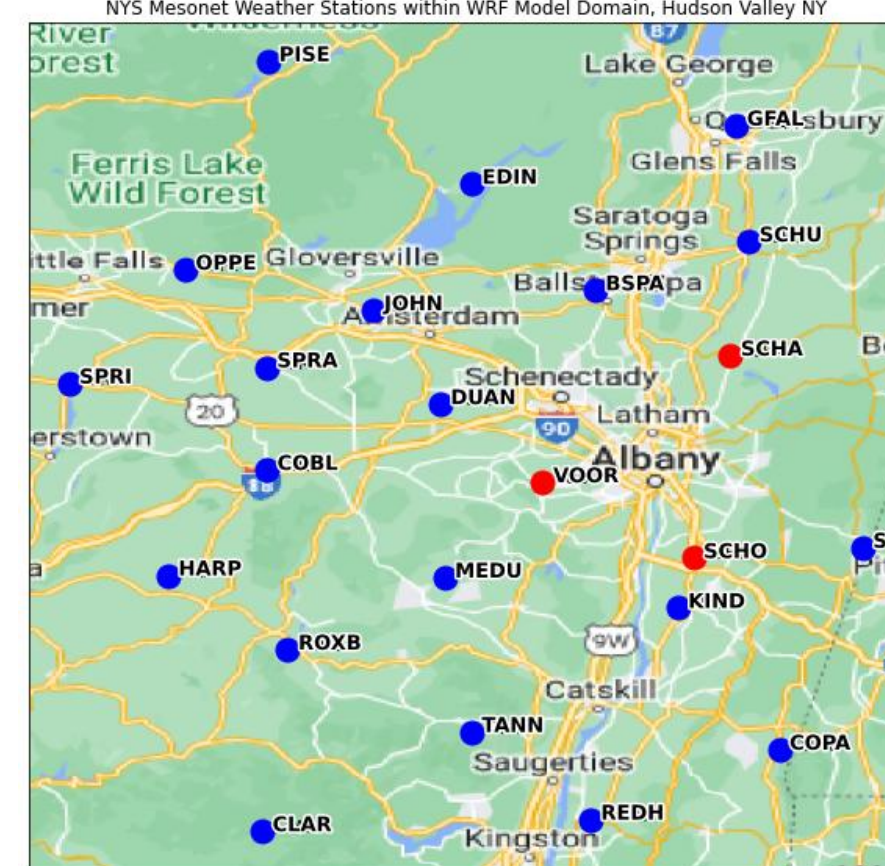


Figure 3. NYS Mesonet Surface Weather Stations in the Hudson Valley, NY, (Google Maps Street Map).

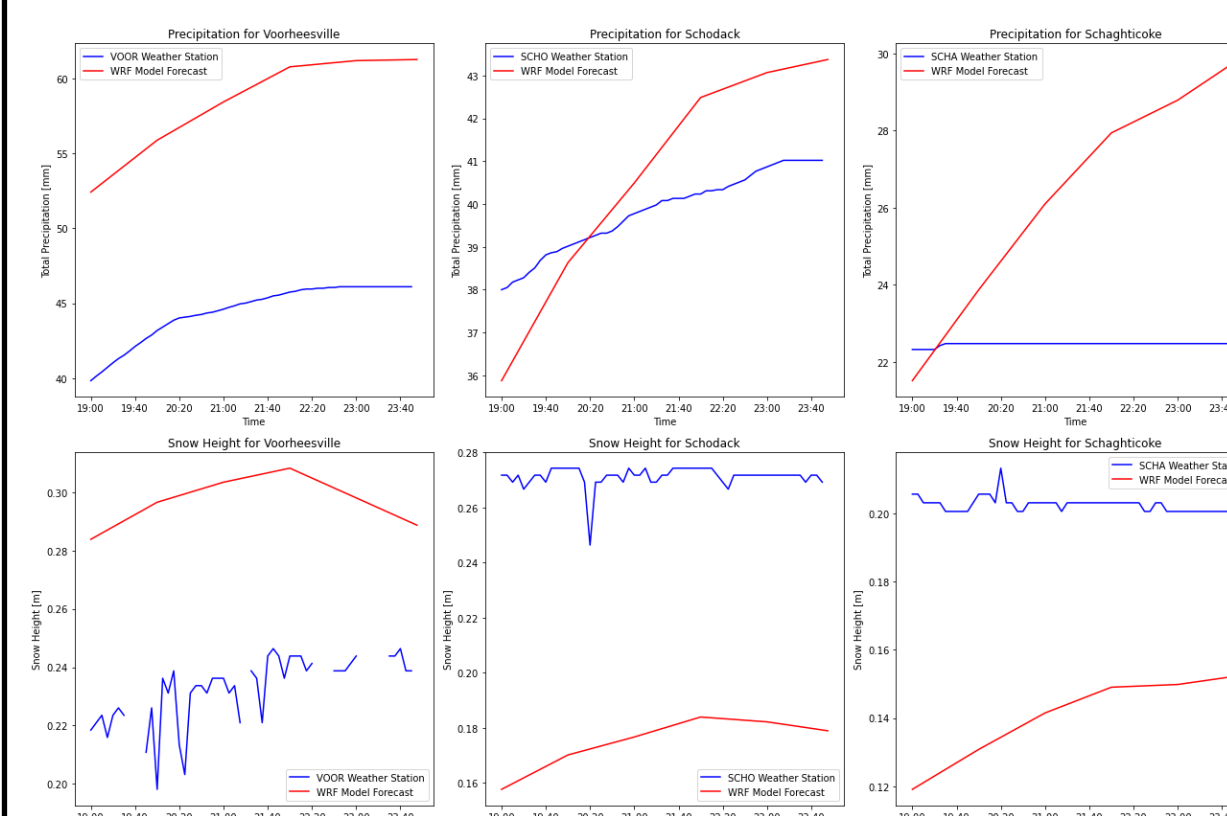


Figure 4. Snow height (meters) and total precipitation (mm) from WRF model output and selected NYS Mesonet surface weather station observations on March 2, 2018, 19:00 UTC to 24:00 UTC.

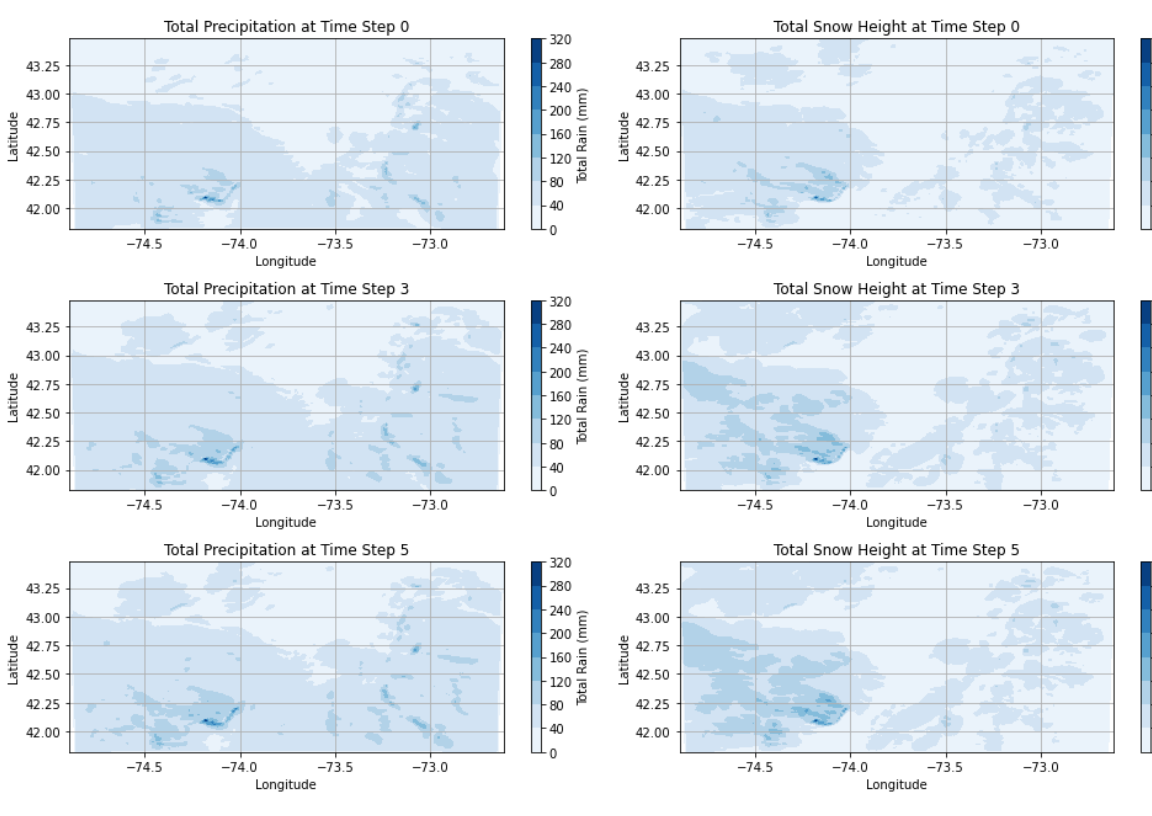


Figure 5. Map of snow height (meters) and total precipitation (mm) from WRF output at time steps 0, 3, & 6.

- As seen in Figure 4., the observations vary dramatically from the model forecast.
- The forecast overpredicted the total amount of precipitation and underpredicted the snow height, with the exception of Voorheesville Station
 - Based on figure 5., it can be assumed that elevation was the deciding factor in terms of where snow was predicted to fall.
 - Overestimated the percentage of precipitation that would occur as rain at low altitudes
 - Overestimated the percentage of precipitation that would occur as snow at high altitudes.
 - This explains the discrepancy in Voorheesville Station
- As a result of the storm, CNN reported that over five people lost their lives and over a million customers lost power across the United States.
- According to Aon Benfield's Impact Forecasting, the storm caused multiple billions of dollars of economic loss and damage.
- Underscores the importance of accurately forecasting the weather, especially for predicting precipitation quantity and type.

Methods to Increase Accuracy.

- There are multiple ways to increase the accuracy of NWP forecasts:
 - Improve data assimilation
 - Model output statistics (MOS)
 - Ensemble forecasting
 - Examining parametrization schemes
 - Machine Learning (ML)**
- ML algorithms derive a relationship between variables in a large database to predict a selected variable
 - ML does not need to understand the physical relationships that create atmospheric conditions and instead derives its own equations between variables
 - This non-numerical processing can lower the computational constraint of computing physics partial derivate equations and can better estimate processes that would be parameterized by taking historical data into account.
- An example of a weather forecast model built using machine learning is GraphCast
 - GraphCast uses Graph Neural Networks and with two sets of data (6 hours ago and current weather) can create a forecast for 6 hours into the future.
 - This process can be repeated to forecast up to 10 days ahead.
 - Outperforms the high-resolution ECMWF model on 90% of variables and lead times (Lam, 2023).

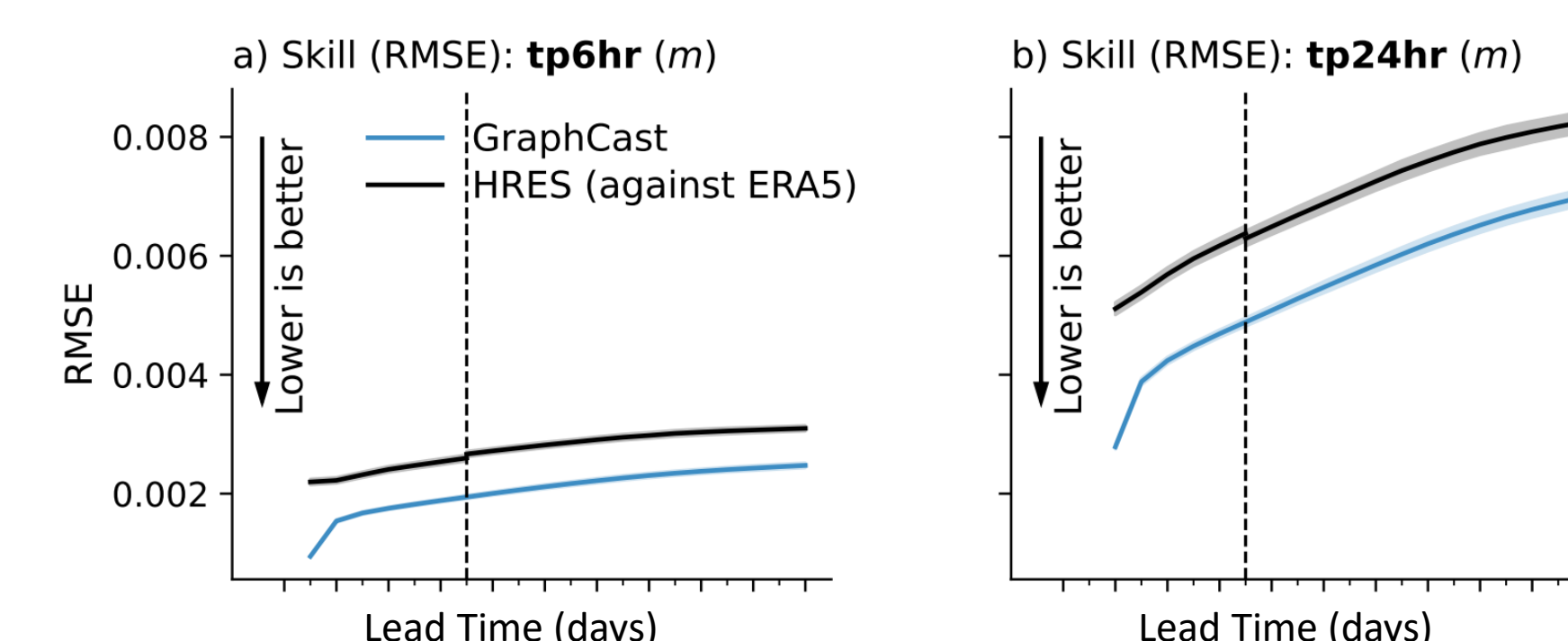


Figure 6. RMSE for Total Precipitation 6 hour (a) & 24 hour (b) GraphCast vs ECMWF HRES in 2018 (Lam 2023) .

Future Work:

- Deep learning (DL) algorithms train input and output nodes in multiple “deep” layers through statistical techniques to identify patterns in data, leading to predictive analysis.
 - These layers form a neural network which can represent non-linear relationships and handle high dimensional data from multiple sources.
 - Can input and output spatial and temporal data.
- NWP model outputs can also be post-processed using DL methods.
 - This allows for the processing of physics-based simulations with historical weather observations and forecasting reanalysis data to keep predictions in line with historical weather.
 - Future work will focus on this topic, using different types of neural networks to increase precipitation accuracy in the Albany region.

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